

Taylor McDonald

taylormcdoc@gmail.com | linkedin.com/in/taylormarie | euvrain.github.io

PROFESSIONAL SUMMARY

Data Science B.S. candidate at NJIT building end-to-end ML systems: NSF-funded climate forecasting at the SETI Institute and self-driven demand forecasting and dynamic pricing platforms using Python, scikit-learn, XGBoost, and FastAPI. Targeting data science and ML engineering roles at tech companies and AI-driven startups.

EDUCATION

New Jersey Institute of Technology

B.S., Data Science.

Newark, NJ

Expected Dec 2027

Relevant Coursework: Intro to ML, Regression Analysis, Statistical Methods I & II, Statistical Methods in Data Science, Multivariate Distribution, Linear Algebra

EXPERIENCE

Machine Learning Research Intern

SETI Institute

Jun 2026 – Present

Remote

- Translated 633 lines of MATLAB implementing DAHD-SL spectral analysis into Python, enabling license-free execution and integration into a GCP-based forecasting workflow.
- Rewrote M-SSA and Varimax Rotation pipeline components for multivariate time series decomposition across 6 spatial channels and 130 time points; authored full inline documentation.
- Collaborated with UCLA and UCSB on an NSF-funded project; presented codebase to Drs. Sudakow and Kondrashov.

Founder

All The Paws LLC

Sep 2025 – Present

East Orange, NJ

- Founded and scaled pet services business from 2 to 15+ clients, maintaining 6-7 regularly rebooking clients; applied scheduling and pricing patterns to directly inform PawsIQ's forecasting and dynamic pricing models.

PROJECTS

PawsIQ: Pet Services Intelligence Platform

In Progress

- Performed EDA on 9,935 NJ bookings; developed GradientBoosting forecasting model achieving 5% RMSE improvement with cross-validation MAE of 0.774 +/- 0.033 across 5 time-series folds.
- Built Ridge Regression pricing model ($R^2=0.727$) producing 16.7% simulated revenue lift over baseline across 1,856 test cases.
- Built React dashboard visualizing demand heatmaps, surge multipliers, and walker performance metrics across 8 active walkers.

Marvel Rivals Tier List Predictor

github.com/euvrain/marvel-rivals-champ-tier-list-predictor

- Built end-to-end unsupervised ML pipeline consuming a REST API, engineered 6 role-normalized features via GroupBy z-score transforms, and deployed an interactive Streamlit dashboard.

House Prices Prediction

github.com/euvrain/housing-price-predictor-linreg-knn-randomforest

- Trained and compared Linear Regression, KNN, and Random Forest on 1,460 samples, achieving $R^2=0.87$ and RMSE=0.14 with 22% RMSE improvement via cross-validated hyperparameter tuning.

ACTIVITIES & LEADERSHIP

Headstarter AI

Software Engineering Fellow

Jul 2024 – Sep 2024

- Built and deployed an AI chatbot and landing page as part of a competitive fellowship focused on AI product development.

Onyx Dance Crew, NJIT

Member and Choreographer

2025 – Present

- Managed logistics for a 40-member collegiate dance crew; choreographed routines for campus performances.

CERTIFICATIONS

- Supervised Machine Learning: Regression and Classification
- Linear Algebra for Machine Learning and Data Science

SKILLS

Technical: Python, scikit-learn, XGBoost, pandas, numpy, SQL, FastAPI, Streamlit, React, Git, matplotlib, seaborn

Professional: Data Storytelling, Technical Documentation, Cross-functional Collaboration, Intermediate Korean